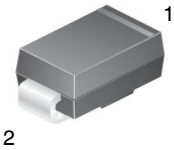
Super Fast Surface Mount Rectifiers

US2AA-US2MA



Cathode Anode

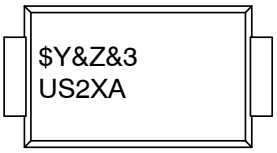


SMA
CASE 403AE

Features

- Glass Passivated Chip Junction
- High Surge Capacity
- Low Forward Voltage Drop
- Fast Switching with Reverse Recovery Time: 50~75 ns Maximum
- UL Flammability 94 V – 0 Classification
- MSL 1 per J–STD–020
- RoHS Compliant / Green Molding Compound
- NRV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC–Q101 Qualified and PPAP Capable*

MARKING DIAGRAM



$Y	= **onsemi** Logo
&Z	= Assembly Plant Code
&3	= Data Code (Year & Week)
US2XA	= Specific Device Code
X	= A/B/D/F/G/J/K/M

ORDERING INFORMATION

See detailed ordering, marking and shipping information on page 1 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 1.

Table 1. ORDERING INFORMATION

Part Number	Top Mark	Package	Packing Method
US2AA	US2AA	DO–214AC (SMA)	Tape and Reel
US2BA	US2BA	DO–214AC (SMA)	Tape and Reel
US2DA	US2DA	DO–214AC (SMA)	Tape and Reel
US2FA, NRVUS2FA*	US2FA	DO–214AC (SMA)	Tape and Reel
US2GA, NRVUS2GA*	US2GA	DO–214AC (SMA)	Tape and Reel
US2JA, NRVUS2JA*	US2JA	DO–214AC (SMA)	Tape and Reel
US2KA, NRVUS2KA*	US2KA	DO–214AC (SMA)	Tape and Reel
US2MA, NRVUS2MA*	US2MA	DO–214AC (SMA)	Tape and Reel

DISCONTINUED (Note 1)

NRVUS2AA*	US2AA	DO–214AC (SMA)	Tape and Reel
NRVUS2BA*	US2BA	DO–214AC (SMA)	Tape and Reel
NRVUS2DA*	US2DA	DO–214AC (SMA)	Tape and Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

1. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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Table 2. ABSOLUTE MAXIMUM RATINGS

Symbol	Parameter	US2 AA	US2 BA	US2 DA	US2 FA	US2 GA	US2 JA	US2 KA	US2 MA	Unit
V_{RRM}	Repetitive Peak Reverse Voltage	50	100	200	300	400	600	800	1000	V
V_{RMS}	RMS Reverse Voltage	35	70	140	210	280	420	560	700	V
V_{DC}	DC Blocking Voltage	50	100	200	300	400	600	800	1000	V
$I_{F(AV)}$	Average Forward Rectified Current	1.5								A
I_{FSM}	Peak Forward Surge Current, 8.3 ms Single Half−Sine Wave, Superimposed on Rated Load	50								A
T_J	Operating Junction Temperature Range	−55 to +150								°C
T_{STG}	Storage Temperature Range	−55 to +150								°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

Table 3. THERMAL CHARACTERISTICS (NOTE 1) (Values are at T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Typical Thermal Resistance, Junction−to−Ambient	189	°C/W
Ψ_{JL}	Typical Thermal Characteristics, Junction−to−Lead (with Reference to Cathode Pin)	31	°C/W

2. Device mounted at minimum pad.

Table 4. ELECTRICAL CHARACTERISTICS (Values are at T_A = 25°C unless otherwise noted)

Symbol	Parameter		Value								Unit
			US2 AA	US2 BA	US2 DA	US2 FA	US2 GA	US2 JA	US2 KA	US2 MA	
V_F	Maximum Instantaneous Forward Voltage (Note2) at Rated $I_{F(AV)}$		1.0				1.3	1.7			V
I_R	Maximum Reverse Current at Rated V_R	TJ = 25°C	5								μA
		TJ = 125°C	100								
t_{rr}	Maximum Reverse Recovery Time (Note 3)		50				75				ns
C_J	Typical Junction Capacitance (Note 4)		50				30				pF

3. Pulse test with PW = 300 μs, 1% duty cycle
4. Reverse recovery test conditions: I_F = 0.5 A, I_R = 1.0 A, I_{RR} = 0.25 A
5. Measured at 1 Mhz and applied reverse voltage of 4.0 V D.C.

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TYPICAL PERFORMANCE CHARACTERISTICS

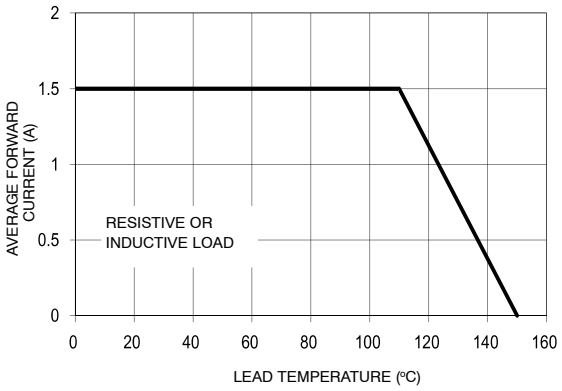


Figure 1. Forward Current Derating Curve

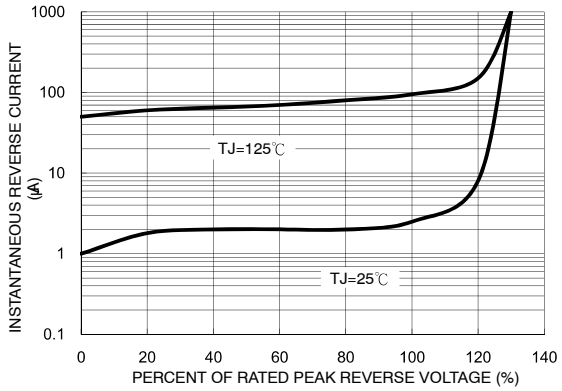


Figure 2. Typical Reverse Characteristics

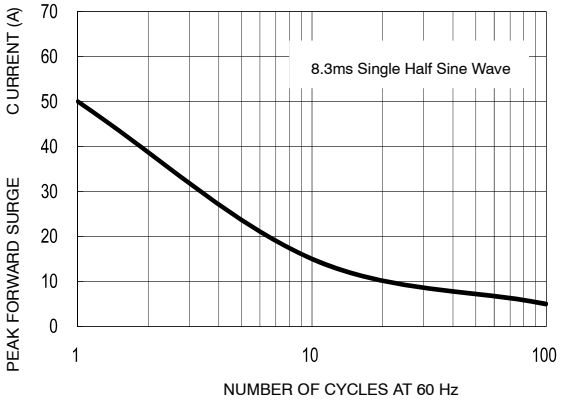


Figure 3. Maximum Non−Repetitive Forward Surge Current

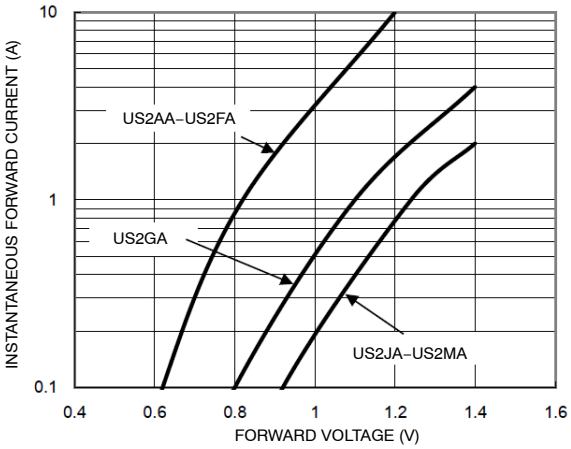


Figure 4. Typical Forward Characteristics

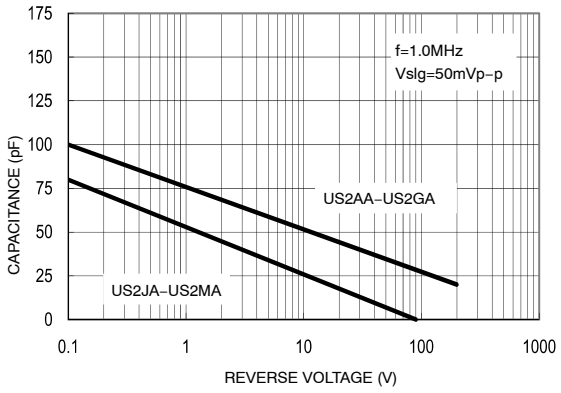


Figure 5. Typical Forward Characteristics

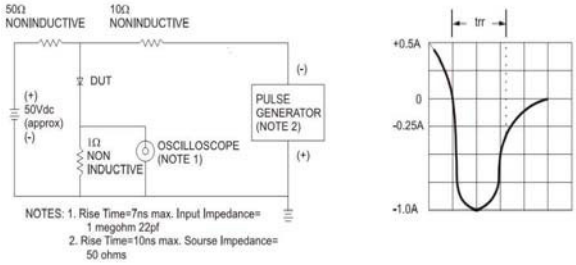
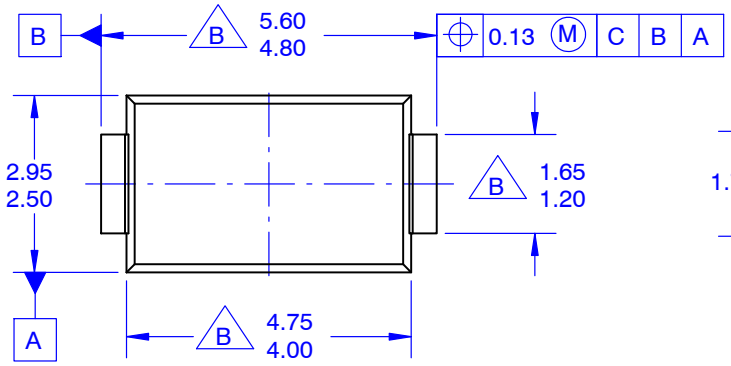


Figure 6. Typical Forward Characteristics

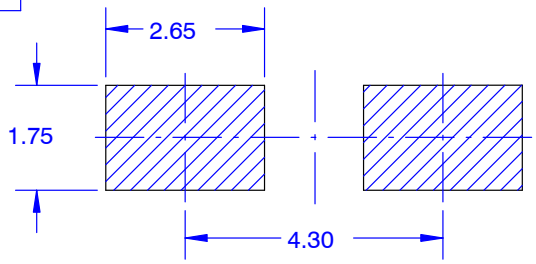
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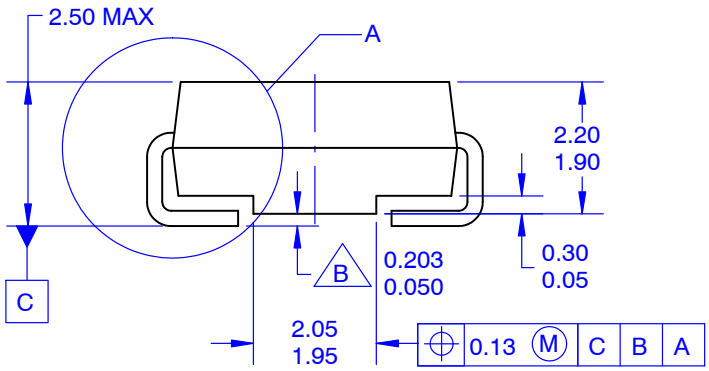
DATE 31 AUG 2016



TOP VIEW



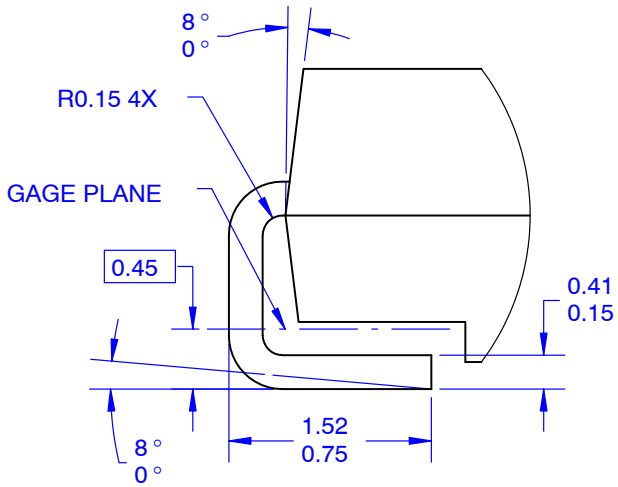
LAND PATTERN RECOMMENDATION



SIDE VIEW

NOTES:

A. EXCEPT WHERE NOTED, CONFORMS TO JEDEC DO214 VARIATION AC.
B DOES NOT COMPLY JEDEC STANDARD VALUE.
C. ALL DIMENSIONS ARE IN MILLIMETERS.
D. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH AND TIE BAR PROTRUSIONS.
E. DIMENSIONS AND TOLERANCE AS PER ASME Y14.5–2009.
E. LAND PATTERN STD. DIOM5025X231M



DETAIL A
SCALE 20 : 1

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